Roll No. .....

TBT–401

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

BIOPHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Discuss the BEER LAMBERT law and write the rules for quantitative. (CO1)

OR

(b) Discuss the principle of spectroscopy and Explain key parameters of spectroscopy. (CO1)

OR

(c) Explain Fluorescence through Jablonski diagram and procedure for collecting a Fluorescence Spectrum. (CO1)

2. (a) Discuss the relationship between IIOMO-LUMO gap and absorbed wavelength of light in UV-Vis Spectroscopv. What is the significance of HOMO-LUMO gap in absorption spectra ? (CO2)

OR

(b) Explain the basic components of spectroscopy and its different types. (CO2)

OR

P. T. O.

(c) Explain the spectroscopy based on Electronic, vibrational and Rotatonal energy levels. (CO2)

3. (a) Explain the Free and Activation Energy. Also discuss with suitable figure for Enzymes lower the activation energy of the reaction but do not change the free energy of the reaction. (CO3)

OR

(b) Derive the expression for Work of Expansion against Constant Pressure and discuss the bomb Calorimtery. (CO3)

(c) Explain the following terms : (CO3)

(i) Thermodynamics of passive and active transport

(ii) Glycolysis

(iii) Negative Entropy processes as applicable for biological systems

4. (a) Explain the intermolecular forces and Dipole-dipole interaction. (CO4)

OR

(b) Explain L-S and j-j coupling with suitable examples. (CO4)

OR

(c) Discuss the Vector Model of Atom and explain the total angular momentum (J_z) of electron. (CO4)

5. (a) Explain the types of isomerism and discuss the structural isomerism with examples. (CO5)

OR

(b) Describe the importance of Quantum Chemistry, and explain the Born Oppenheimer Approximation. (CO5)

OR

(c) Explain the Wave Function Theory and Derive Hartree-Fock Theory. (CO5)

Roll No. .....

TBT–402

B. TECH. (BIOTECHNOLOGY) (FOURTH SEMESTER)

END SEMESTER EXAMINATION, June, 2024

IMMUNOLOGY AND IMMUNOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any ***two*** sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What is primary and secondary immune response ? Explain the role of primary immune organs in acquired immune response. (CO1)

   (b) What is inflammation ? Explain the role of chemotactic factors in eliciting inflammatory response during injury. (CO1)

   (c) Explain the mechanism of activation of B-cell mediated Immune response. (CO2)

2. (a) Explain the contribution of Louis Pasteur in the field of immunology. (CO1)

   (b) What is RIA ? Discuss their significance in immune-diagnosis of a cancer. (CO4)

   (c) What are vaccines ? Describe the mechanism of synthesis of recombinant DNA vaccines. (CO4)

P. T. O.

3. (a) What is MAC ? Explain the Classical pathway of complement proteins forming MAC against bacterial pathogen. (CO2)

   (b) What are the forces that exists during an antigen and antibody interaction ? Discuss the Ag-Ab interactions in detail. (CO3)

   (c) Using suitable diagrams, differentiate the types of MHCs and their role in antigen presentation. (CO3)

4. (a) What is agglutination ? With reference to antibody titer, how you can confirm TYPHOID infection using bacterial agglutination test ? (CO4)

   (b) Comment on any *two* of the following : (CO4)

      (i) Immuno-electrophoresis

      (ii) Immuno-precipitation

      (iii) CFT

   (c) A patient is suspected to be suffering from Hepatitis B virus ? Using diagrams, explain the ELISA techniques you will perform to detect anti-HBs in the patient's serum ? (CO4)

5. (a) What are transplant antigens ? Explain the types of transplant antigens and the associated immune response involved in the rejection of a transplant. (CO5)

   (b) Explain the role of immune complexes in inducing Type-I hypersensitivity reaction in human body. (CO5)

   (c) What are tumor specific antigens ? With suitable examples, explain immune response against tumor antigens. (CO5)

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End Semester Examination 2023-24

Name of the Course: B.Tech. Biotechnology — Semester: IV

Name of the Paper: Bio-Analytical Techniques — Paper Code: **TBT-403**

Time: 3 Hours — Maximum Marks: 100

Note:-

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks

Q1	(20marks)	**CO 1**
(a)	Discuss instrumentation, working and application of fluorescence microscopy?	
(b)	Discuss instrumentation, working and application of AFM?	
(c)	Discuss instrumentation, working and application of TEM?	
Q2	(20 marks)	**CO 2**
(a)	Write short note on (i). Iso electric focusing (ii). PAGE	
(b)	(i). Discuss various factor effecting centrifugation process with the help of equation (ii). Draw a relationship between different forces acting during centrifugation process?	
(c)	Discuss at least 10 industrial applications of centrifugation and electrophoresis process	
Q3	(20marks)	**CO 3**
(a)	Discuss instrumentation, working, graph analysis and application of Gas chromatography?	
(b)	Discuss instrumentation, working, application of HPLC? Discuss major difference between normal and reverse HPLC	
(c)	Discuss instrumentation, working, graph analysis and application of size exclusion chromatography?	
Q4	(20marks)	**CO 4**
(a)	Discuss working of UV visible spectroscopy? How spectroscopy is utilized for protein quantification?	
(b)	Discuss instrumentation, working and application of Mass Spectroscopy?	
(c)	Discuss and derive Beer Lambert Law?	
Q5	(20marks)	**CO 5**
(a)	Illustrate working of RIA in detail along with its application, advantages and disadvantages?	
(b)	(i). Differentiate between different forms of ionizing radiation (ii). Write short note on different unit to measure radioactivity?	
(c)	Discuss different methods to measure radioactivity?	

Even End Semester Examination 2024

Name of the Program: **B.Tech. Biotechnology** *Semester:* **4th**

Name of the Course: **Environmental Biotechnology** *Course Code:* **TBT404**

Time: **3hrs** Maximum Marks: **100**

Note:

i. Answer **all the questions** by choosing ***any two of the sub questions***.

ii. Each question carries 20 marks

Q1	(20 marks)	**CO4 & CO5**
(a)	Highlight the role of degradation mechanism of xenobiotics with example.	
(b)	Diagrammatically explain the photo-catalysis process mediated byTitanium dioxide particles	
(c)	Discuss the strategies to be followed for restoration of waste lands.	
Q2	(20marks)	**CO3 & CO5**
(a)	Differentiate between bio-catalysis and bio-transformation with examples.	
(b)	"*A tool used to assess the significant effects of a project ordevelopment proposal on the environment*" Comment on the afore mentioned statement highlight its objectives.	
(c)	What are genetically modified organisms? Explain the negative impactsimposed by GMOs on health and the environment.	
Q3	(20 marks)	**CO5 & CO2**
(a)	Explain the techniques followed for the bioremediation of toxic metal from the environment.	
(b)	Differentiate between Membrane bed reactor and Up-flow anaerobic sludge blanket reactor.	
(c)	Describe the construction design of a sanitary landfill.	
Q4	(20 marks)	**CO4, CO5 & CO3**
(a)	Explain the mechanism for biotransformation of pesticides.	
(b)	Explain the role of Government officials for the disposal of solidwastes.	
(c)	"*Exposure of sewage sludge or other types of wet organic waste tohigh temperature and pressure*" Explain such process in detail.	
Q5	(20 marks)	**CO5, CO2 & CO1**
(a)	Explain the microbe assisted method for the degradation of plastics.	
(b)	Discuss the after effects caused by Ozone layer depletion with examples.	
(c)	Discuss the applications of any five enzymes which are effective in thecleaning of environmental remnants.	

Even End Semester Examination 2024

Name of the Program: **B.Tech. Biotechnology** *Semester:* **4th**
Name of the Course: **Waste Management & Upcycling** *Course Code:* **TBT405a**

Time: **3hrs** Maximum Marks: **100**

Note:

i. Answer **all the questions** by choosing ***any two of the sub questions.***
ii. Each question carries 20 marks

Q1	(20 marks)	**CO1**
(a)	Give detailed account of global waste generation and EU classification of the waste.	
(b)	Explain the process of waste water treatment with the help of labelled diagram.	
(c)	Discuss the Processes of Gasification, Incineration and Pyrolysis	
Q2	(20 marks)	**CO3**
(a)	What is vermicomposting and how it helps in waste upcycling? Provide any case study in support of your answer.	
(b)	Draw a labelled diagram of engineered landfill site for managing hazardous waste.	
(c)	Mention the name and impacts of new and emerging air pollutants and mention how air pollution control devices help in removal of air pollutants.	
Q3	(20 marks)	**CO2**
(a)	What is biomedical waste and how it is managed through color coded scheme.	
(b)	Discuss the health considerations in the context of operating facilities?	
(c)	What is landfill leaching and how does land fill engineering helps in landfill management?	
Q4	(20 marks)	**CO4**
(a)	How polluted lands can be used for sustainable of crop production. Give examples in support of your answer?	
(b)	What are the advances in waste recycling and delivery technologies to deliver value added products?	
(c)	Write short notes on i) carbon foot print ii) life cycle analysis	
Q5	(20 marks)	**CO5**
(a)	How organic waste can be upcycled into different useful products. Provide suitable examples in support of your answer.	

	Explain in detail the concepts of upcycling, recycling and down cycling.	
(b)	How wasteland can be a sustainable solution for bioenergy production what are the challenges and opportunities?	
(c)	What is Landfill mining and how does it help in solid waste management?	

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EVEN END SEMESTER EXAMINATION -2024

Name of the Course: B.Tech. Biotech ***Semester:*** *IV*

Name of the Paper: Biomaterial ***Course Code:*** TBT 405 C

Time: 03 Hours Maximum Marks: 100

Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

Q1	**(20 marks)**	
(a)	Write a short note on the following: (a) shape memory alloy, (b) Elastin fiber, (c) vasodilation, and (d) strain recovery with examples.	**CO1,**
(b)	Draw and explain the stress-strain graph tendon and human abdominal skin. Compare the mechanical properties of the same.	**CO1,**
(c)	Differentiate between surface erosion and bulk erosion with suitable example. What are the factors that influence bulk erosion?	**CO2**
Q2	**(20 marks)**	
(a)	What is autocatalysis? Explain types of degradation. Any four.	**CO2,**
(b)	What are the factors that supports formation of biofilm. Lists any four biomaterials along with their environment for different implants of human body.	**CO2,**
(c)	Explain the following methods of tool characterization: (a) Thermogravimetric analysis, (b) FTIR, (c) DSC, and (d) MALDI	**CO3**
Q3	**(20 marks)**	**CO3,**
(a)	What is biocompatibility and cytotoxicity? how cytotoxicity assessed in in-vitro? Explain in brief	**CO4,**
(b)	What are the different types of corrosion occurred in metallic biomaterial? Explain in brief.	**CO3**
(c)	What are the different methods of preventing corrosion in metals. Explain any three.	
Q4	**(20 marks)**	
(a)	What are composites? Explain any three types of composites. what are the advantages of composite over polymer. List any five.	**CO4,**
(b)	How natural fibers are different from synthetic fibers. Give your justification.	**CO4,**
(c)	What are the reasons of blending different polymers. Explain any three methods of blending of polymer.	**CO2**
Q5	**(20 marks)**	
(a)	Write a short note on natural protein. Explain any four natural proteins applied in biomaterial.	**CO4,**
(b)	Explain the process of drawing with neat sketch.	**CO5,**
(c)	Elaborate the metal casting and extrusion in brief.	**CO5**

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